COMPUTER ENGINEERING

QUEZON CITY

COURSE CODE/TITLE:	CPE 212- Automating Server Management	**SECTION:**	CPE31S2
DATE PERFORMED:	September 1, 2026	**DATE SUBMITTED:**	September 1, 2026
NAME:	Nadine A. Malabanan	**INSTRUCTOR:**	Engr. Jimlord Queado

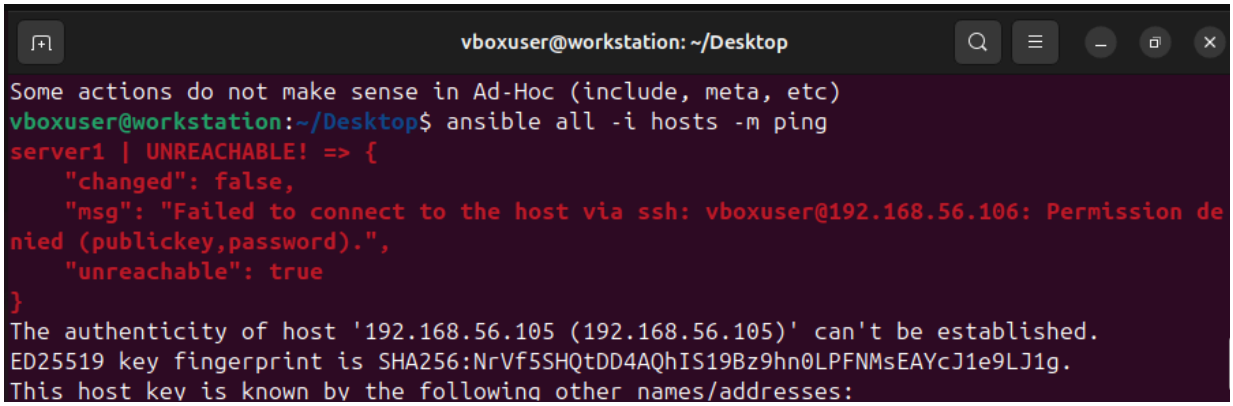
ACTIVITY NO. __4__ Hands-on Activity: Ansible Basics

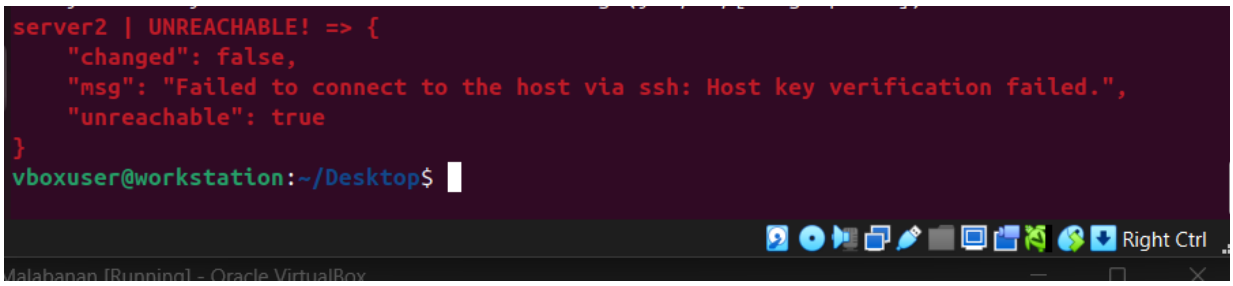
1. PROCEDURE OUTPUT

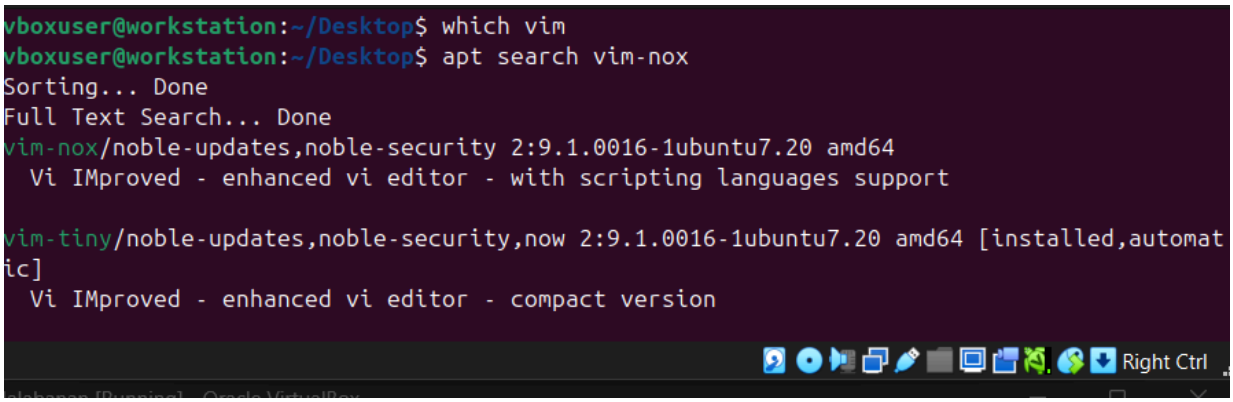
What procedures did you perform?

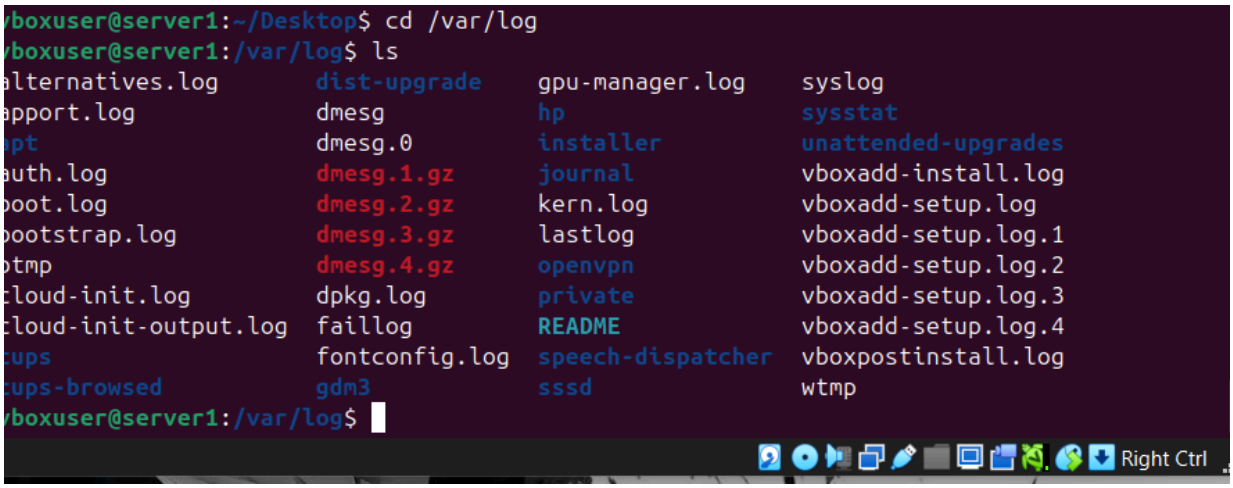
TASK 1

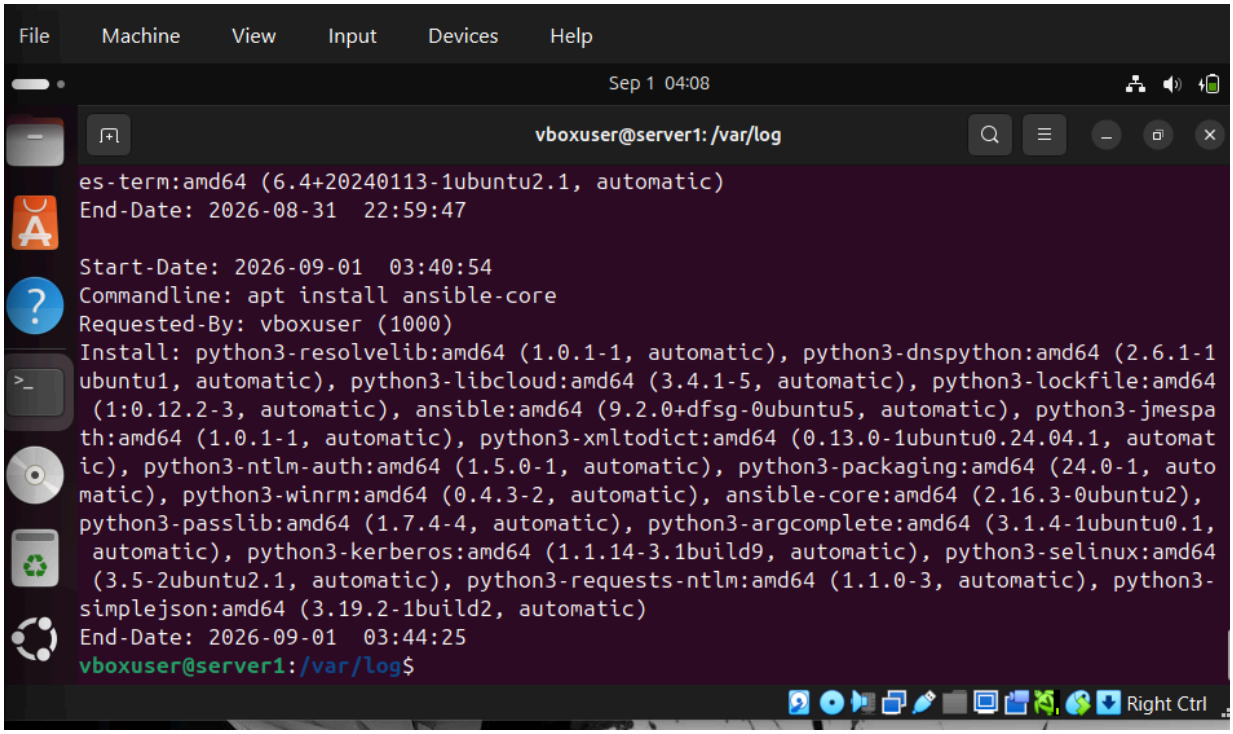
```
Selecting previously unselected package python3-passlib.
Preparing to unpack .../16-python3-passlib_1.7.4-4_all.deb ...
Unpacking python3-passlib (1.7.4-4) ...
Setting up python3-lockfile (1:0.12.2-3) ...
Setting up python3-passlib (1.7.4-4) ...
Setting up python3-ntlm-auth (1.5.0-1) ...
Setting up python3-resolvelib (1.0.1-1) ...
Setting up python3-kerberos (1.1.14-3.1build9) ...
Setting up python3-simplejson (3.19.2-1build2) ...
Setting up python3-xmltodict (0.13.0-1ubuntu0.24.04.1) ...
Setting up python3-packaging (24.0-1) ...
Setting up python3-jmespath (1.0.1-1) ...
Setting up python3-dnspython (2.6.1-1ubuntu1) ...
Setting up python3-selinux (3.5-2ubuntu2.1) ...
Setting up python3-argcomplete (3.1.4-1ubuntu0.1) ...
Setting up python3-requests-ntlm (1.1.0-3) ...
Setting up ansible-core (2.16.3-0ubuntu2) ...
Setting up python3-libcloud (3.4.1-5) ...
Setting up python3-winrm (0.4.3-2) ...
Setting up ansible (9.2.0+dfsg-0ubuntu5) ...
Processing triggers for man-db (2.12.0-4build2) ...
vboxuser@workstation:~$
```

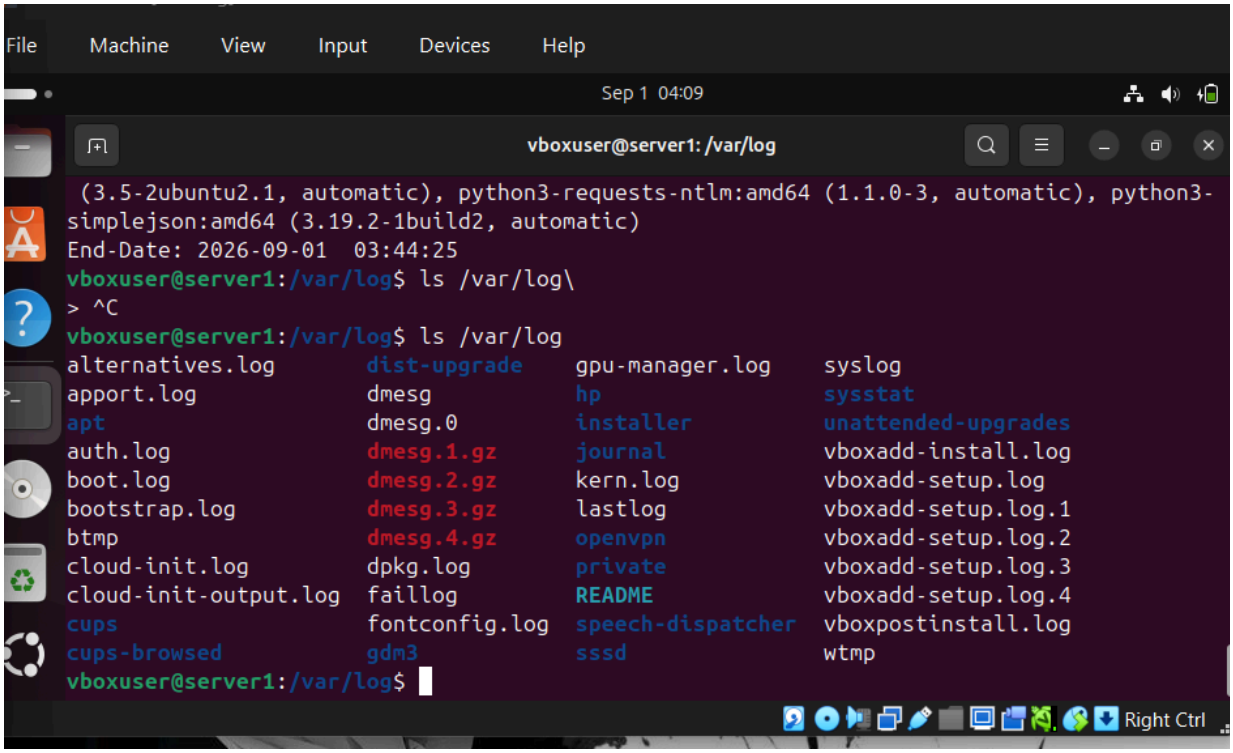

vboxuser@workstation: ~/Desktop
Some actions do not make sense in Ad-Hoc (include, meta, etc)
vboxuser@workstation:~/Desktop$ ansible all -i hosts -m ping
server1 | UNREACHABLE! => {
"changed": false,
"msg": "Failed to connect to the host via ssh: vboxuser@192.168.56.106: Permission de
nied (publickey,password).",
"unreachable": true
}
The authenticity of host '192.168.56.105 (192.168.56.105)' can't be established.
ED25519 key fingerprint is SHA256:NrVf5SHQtDD4AQhIS19Bz9hn0LPFNMsEAYcJ1e9LJ1g.
This host key is known by the following other names/addresses:

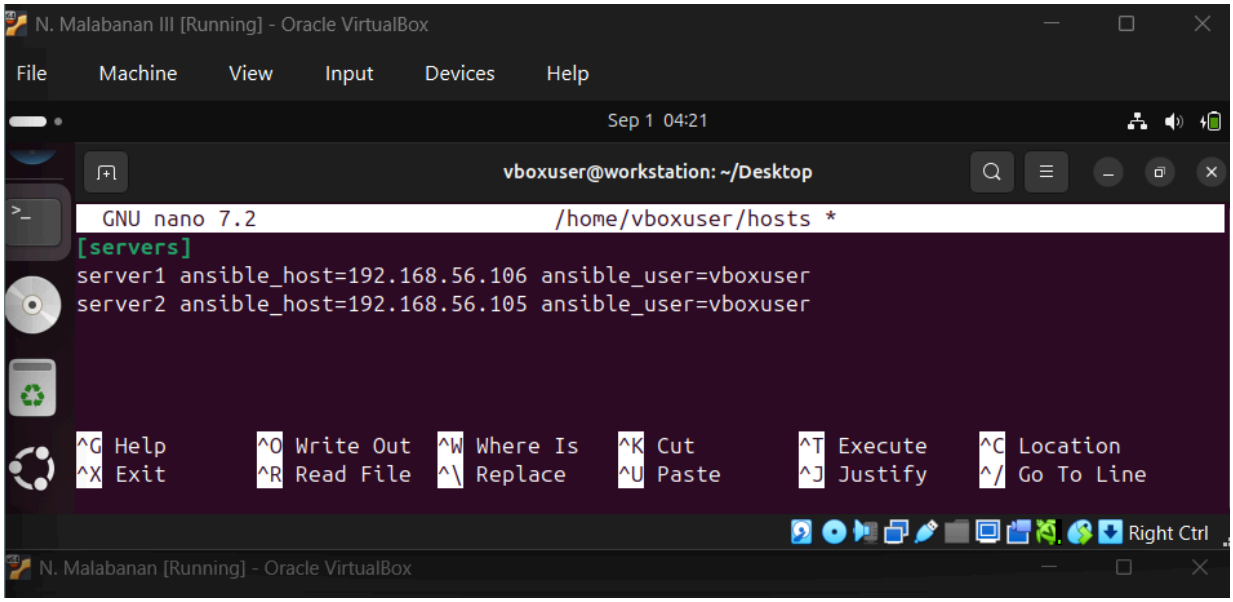

server2 | UNREACHABLE! => {
"changed": false,
"msg": "Failed to connect to the host via ssh: Host key verification failed.",
"unreachable": true
}
vboxuser@workstation:~/Desktop$
Right Ctrl

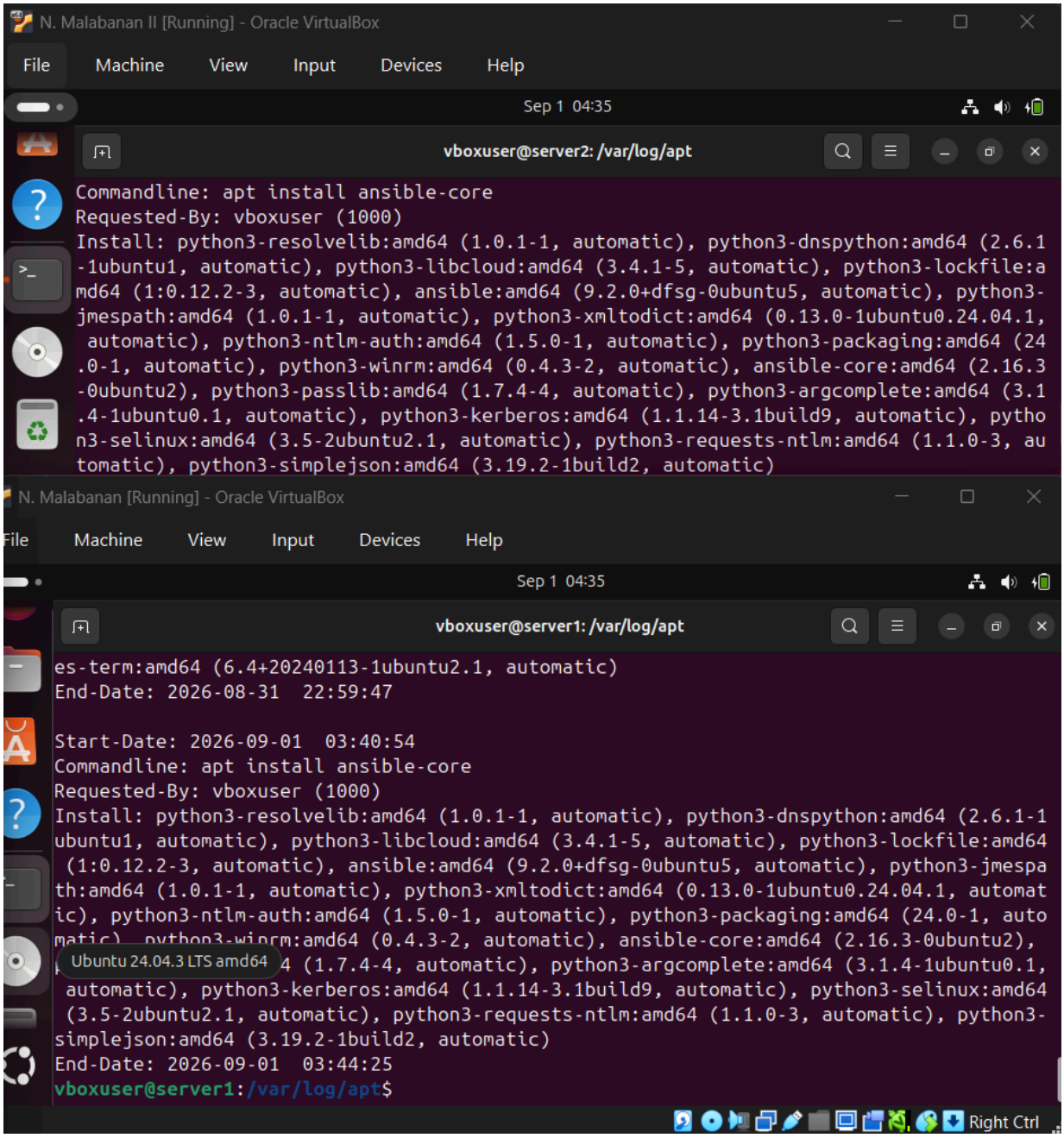

vboxuser@workstation:~/Desktop$ which vim
vboxuser@workstation:~/Desktop$ apt search vim-nox
Sorting... Done
Full Text Search... Done
vim-nox/noble-updates,noble-security 2:9.1.0016-1ubuntu7.20 amd64
Vi IMproved - enhanced vi editor - with scripting languages support
vim-tiny/noble-updates,noble-security,now 2:9.1.0016-1ubuntu7.20 amd64 [installed,automat
ic]
Vi IMproved - enhanced vi editor - compact version
Right Ctrl

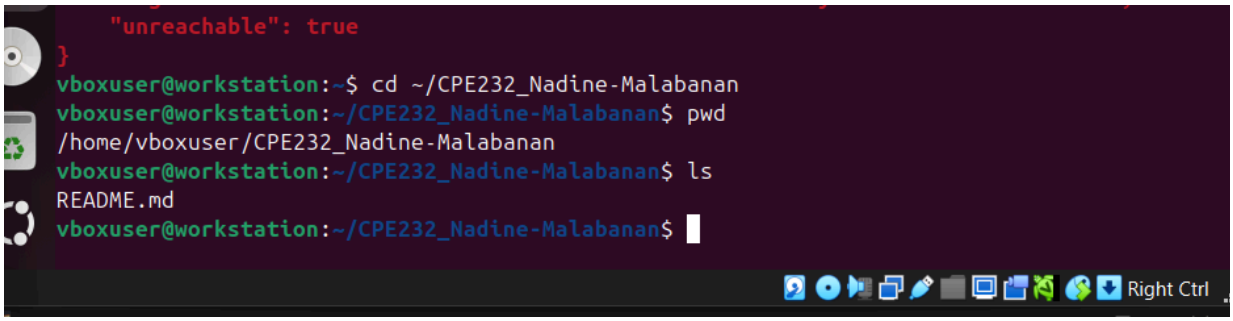

vboxuser@server1:~/Desktop$ cd /var/log
vboxuser@server1:/var/log$ ls
alternatives.log
apport.log
apt
auth.log
boot.log
bootstrap.log
btmp
cloud-init.log
cloud-init-output.log
cups
cups-browsed
dist-upgrade
dmesg
dmesg.0
dmesg.1.gz
dmesg.2.gz
dmesg.3.gz
dmesg.4.gz
dpkg.log
faillog
fontconfig.log
gdm3
gpu-manager.log
hp
installer
journal
kern.log
lastlog
openvpn
private
README
speech-dispatcher
sssd
syslog
sysstat
unattended-upgrades
vboxadd-install.log
vboxadd-setup.log
vboxadd-setup.log.1
vboxadd-setup.log.2
vboxadd-setup.log.3
vboxadd-setup.log.4
vboxpostinstall.log
wtmp
vboxuser@server1:/var/log$
Right Ctrl


File Machine View Input Devices Help
Sep 1 04:08
vboxuser@server1: /var/log
es-term:amd64 (6.4+20240113-1ubuntu2.1, automatic)
End-Date: 2026-08-31 22:59:47
Start-Date: 2026-09-01 03:40:54
Commandline: apt install ansible-core
Requested-By: vboxuser (1000)
Install: python3-resolvelib:amd64 (1.0.1-1, automatic), python3-dnspython:amd64 (2.6.1-1 ubuntu1, automatic), python3-libcloud:amd64 (3.4.1-5, automatic), python3-lockfile:amd64 (1:0.12.2-3, automatic), ansible:amd64 (9.2.0+dfsg-0ubuntu5, automatic), python3-jmespath:amd64 (1.0.1-1, automatic), python3-xmltodict:amd64 (0.13.0-1ubuntu0.24.04.1, automatic), python3-ntlm-auth:amd64 (1.5.0-1, automatic), python3-packaging:amd64 (24.0-1, automatic), python3-winrm:amd64 (0.4.3-2, automatic), ansible-core:amd64 (2.16.3-0ubuntu2), python3-passlib:amd64 (1.7.4-4, automatic), python3-argcomplete:amd64 (3.1.4-1ubuntu0.1, automatic), python3-kerberos:amd64 (1.1.14-3.1build9, automatic), python3-selinux:amd64 (3.5-2ubuntu2.1, automatic), python3-requests-ntlm:amd64 (1.1.0-3, automatic), python3-simplejson:amd64 (3.19.2-1build2, automatic)
End-Date: 2026-09-01 03:44:25
vboxuser@server1:/var/log$
Right Ctrl

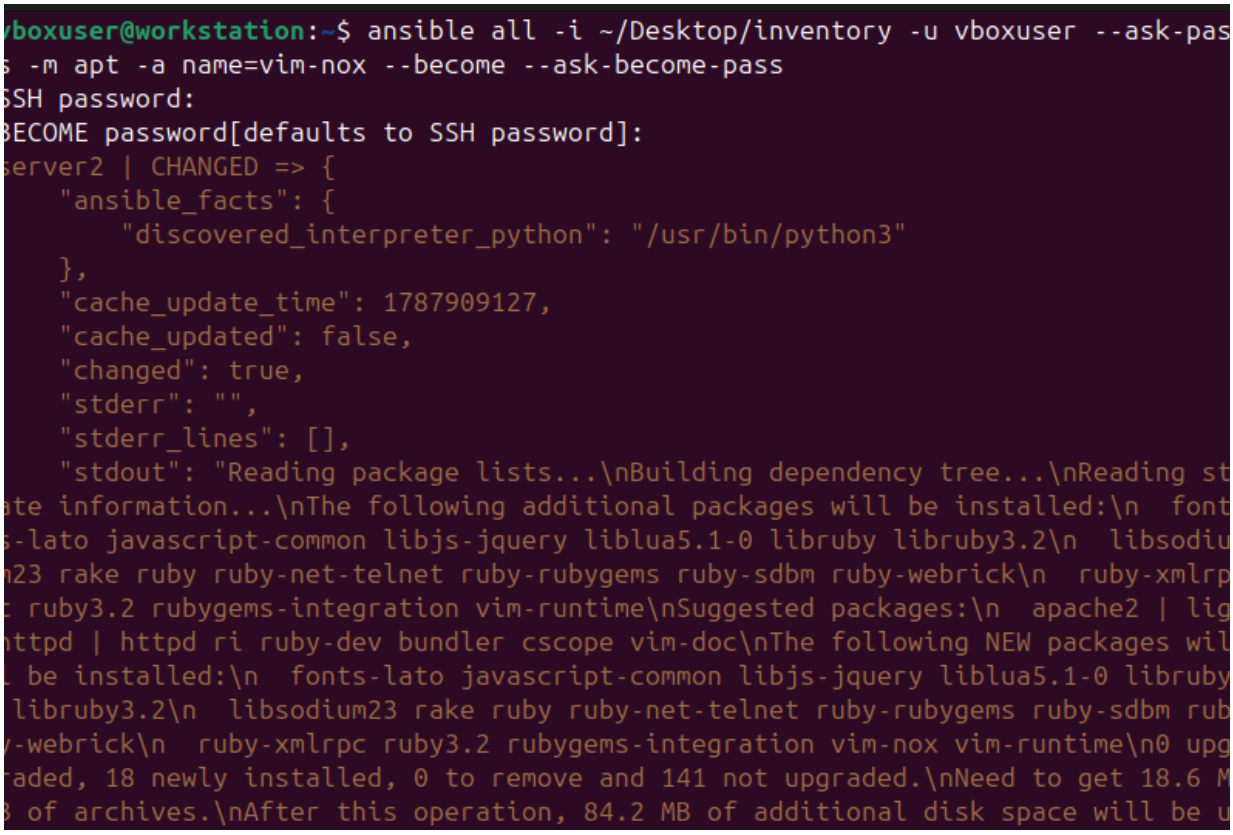

File Machine View Input Devices Help
Sep 1 04:09
vboxuser@server1: /var/log
(3.5-2ubuntu2.1, automatic), python3-requests-ntlm:amd64 (1.1.0-3, automatic), python3-simplejson:amd64 (3.19.2-1build2, automatic)
End-Date: 2026-09-01 03:44:25
vboxuser@server1:/var/log$ ls /var/log\
> ^C
vboxuser@server1:/var/log$ ls /var/log
alternatives.log dist-upgrade gpu-manager.log syslog
apport.log dmesg hp sysstat
apt dmesg.0 installer unattended-upgrades
auth.log dmesg.1.gz journal vboxadd-install.log
boot.log dmesg.2.gz kern.log vboxadd-setup.log
bootstrap.log dmesg.3.gz lastlog vboxadd-setup.log.1
btmp dmesg.4.gz openvpn vboxadd-setup.log.2
cloud-init.log dpkg.log private vboxadd-setup.log.3
cloud-init-output.log faillog README vboxadd-setup.log.4
cups fontconfig.log speech-dispatcher vboxpostinstall.log
cups-browsed gdm3 sssd wtmp
vboxuser@server1:/var/log$
Right Ctrl


N. Malabanan III [Running] - Oracle VirtualBox
File
Machine
View
Input
Devices
Help
Sep 1 04:21
vboxuser@workstation: ~/Desktop
GNU nano 7.2 /home/vboxuser/hosts *
[servers]
server1 ansible_host=192.168.56.106 ansible_user=vboxuser
server2 ansible_host=192.168.56.105 ansible_user=vboxuser
^G Help
^X Exit
^O Write Out
^R Read File
^W Where Is
^\ Replace
^K Cut
^U Paste
^T Execute
^J Justify
^C Location
^/ Go To Line
Right Ctrl
N. Malabanan [Running] - Oracle VirtualBox


N. Malabanan II [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Sep 1 04:35
vboxuser@server2: /var/log/apt
Commandline: apt install ansible-core
Requested-By: vboxuser (1000)
Install: python3-resolvelib:amd64 (1.0.1-1, automatic), python3-dnspython:amd64 (2.6.1-1ubuntu1, automatic), python3-libcloud:amd64 (3.4.1-5, automatic), python3-lockfile:amd64 (1:0.12.2-3, automatic), ansible:amd64 (9.2.0+dfsg-0ubuntu5, automatic), python3-jmespath:amd64 (1.0.1-1, automatic), python3-xmltodict:amd64 (0.13.0-1ubuntu0.24.04.1, automatic), python3-ntlm-auth:amd64 (1.5.0-1, automatic), python3-packaging:amd64 (24.0-1, automatic), python3-winrm:amd64 (0.4.3-2, automatic), ansible-core:amd64 (2.16.3-0ubuntu2), python3-passlib:amd64 (1.7.4-4, automatic), python3-argcomplete:amd64 (3.1.4-1ubuntu0.1, automatic), python3-kerberos:amd64 (1.1.14-3.1build9, automatic), python3-selinux:amd64 (3.5-2ubuntu2.1, automatic), python3-requests-ntlm:amd64 (1.1.0-3, automatic), python3-simplejson:amd64 (3.19.2-1build2, automatic)
N. Malabanan [Running] - Oracle VirtualBox
File Machine View Input Devices Help
Sep 1 04:35
vboxuser@server1: /var/log/apt
es-term:amd64 (6.4+20240113-1ubuntu2.1, automatic)
End-Date: 2026-08-31 22:59:47
Start-Date: 2026-09-01 03:40:54
Commandline: apt install ansible-core
Requested-By: vboxuser (1000)
Install: python3-resolvelib:amd64 (1.0.1-1, automatic), python3-dnspython:amd64 (2.6.1-1ubuntu1, automatic), python3-libcloud:amd64 (3.4.1-5, automatic), python3-lockfile:amd64 (1:0.12.2-3, automatic), ansible:amd64 (9.2.0+dfsg-0ubuntu5, automatic), python3-jmespath:amd64 (1.0.1-1, automatic), python3-xmltodict:amd64 (0.13.0-1ubuntu0.24.04.1, automatic), python3-ntlm-auth:amd64 (1.5.0-1, automatic), python3-packaging:amd64 (24.0-1, automatic), python3-winrm:amd64 (0.4.3-2, automatic), ansible-core:amd64 (2.16.3-0ubuntu2), Ubuntu 24.04.3 LTS amd64 (1.7.4-4, automatic), python3-argcomplete:amd64 (3.1.4-1ubuntu0.1, automatic), python3-kerberos:amd64 (1.1.14-3.1build9, automatic), python3-selinux:amd64 (3.5-2ubuntu2.1, automatic), python3-requests-ntlm:amd64 (1.1.0-3, automatic), python3-simplejson:amd64 (3.19.2-1build2, automatic)
End-Date: 2026-09-01 03:44:25
vboxuser@server1:/var/log/apt$
Right Ctrl


"unreachable": true
}
vboxuser@workstation:~$ cd ~/CPE232_Nadine-Malabanan
vboxuser@workstation:~/CPE232_Nadine-Malabanan$ pwd
/home/vboxuser/CPE232_Nadine-Malabanan
vboxuser@workstation:~/CPE232_Nadine-Malabanan$ ls
README.md
vboxuser@workstation:~/CPE232_Nadine-Malabanan$
Right Ctrl


"unreachable": true
}
vboxuser@workstation:~/Desktop$ ansible all -i ~/Desktop/inventory -u vboxuser --ask-pass -m ping
SSH password:
server1 | SUCCESS => {
"ansible_facts": {
"discovered_interpreter_python": "/usr/bin/python3"
},
"changed": false,
"ping": "pong"
}
server2 | SUCCESS => {
"ansible_facts": {
"discovered_interpreter_python": "/usr/bin/python3"
},
"changed": false,
"ping": "pong"
}
vboxuser@workstation:~/Desktop$


vboxuser@workstation:~$ ansible all -i ~/Desktop/inventory -u vboxuser --ask-pas
s -m apt -a name=vim-nox --become --ask-become-pass
SSH password:
BECOME password[defaults to SSH password]:
server2 | CHANGED => {
"ansible_facts": {
"discovered_interpreter_python": "/usr/bin/python3"
},
"cache_update_time": 1787909127,
"cache_updated": false,
"changed": true,
"stderr": "",
"stderr_lines": [],
"stdout": "Reading package lists...\nBuilding dependency tree...\nReading st
ate information...\nThe following additional packages will be installed:\n font
s-lato javascript-common libjs-jquery liblua5.1-0 libruby libruby3.2\n libsodiu
m23 rake ruby ruby-net-telnet ruby-rubygems ruby-sdbm ruby-webrick\n ruby-xmlrp
c ruby3.2 rubygems-integration vim-runtime\nSuggested packages:\n apache2 | lig
httpd | httpd ri ruby-dev bundler cscope vim-doc\nThe following NEW packages wil
l be installed:\n fonts-lato javascript-common libjs-jquery liblua5.1-0 libruby
libruby3.2\n libsodium23 rake ruby ruby-net-telnet ruby-rubygems ruby-sdbm rub
y-webrick\n ruby-xmlrpc ruby3.2 rubygems-integration vim-nox vim-runtime\n0 upg
raded, 18 newly installed, 0 to remove and 141 not upgraded.\nNeed to get 18.6 M
B of archives.\nAfter this operation, 84.2 MB of additional disk space will be u

```
^C
vboxuser@server1:~/Desktop$ sudo apt install vim-nox
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
vim-nox is already the newest version (2:9.1.0016-1ubuntu7.20).
0 upgraded, 0 newly installed, 0 to remove and 186 not upgraded.
vboxuser@server1:~/Desktop$ which vim
/usr/bin/vim
vboxuser@server1:~/Desktop$ apt search vim-nox
Sorting... Done
Full Text Search... Done
vim-nox/noble-updates,noble-security,now 2:9.1.0016-1ubuntu7.20 amd64 [installed]
  Vi IMproved - enhanced vi editor - with scripting languages support

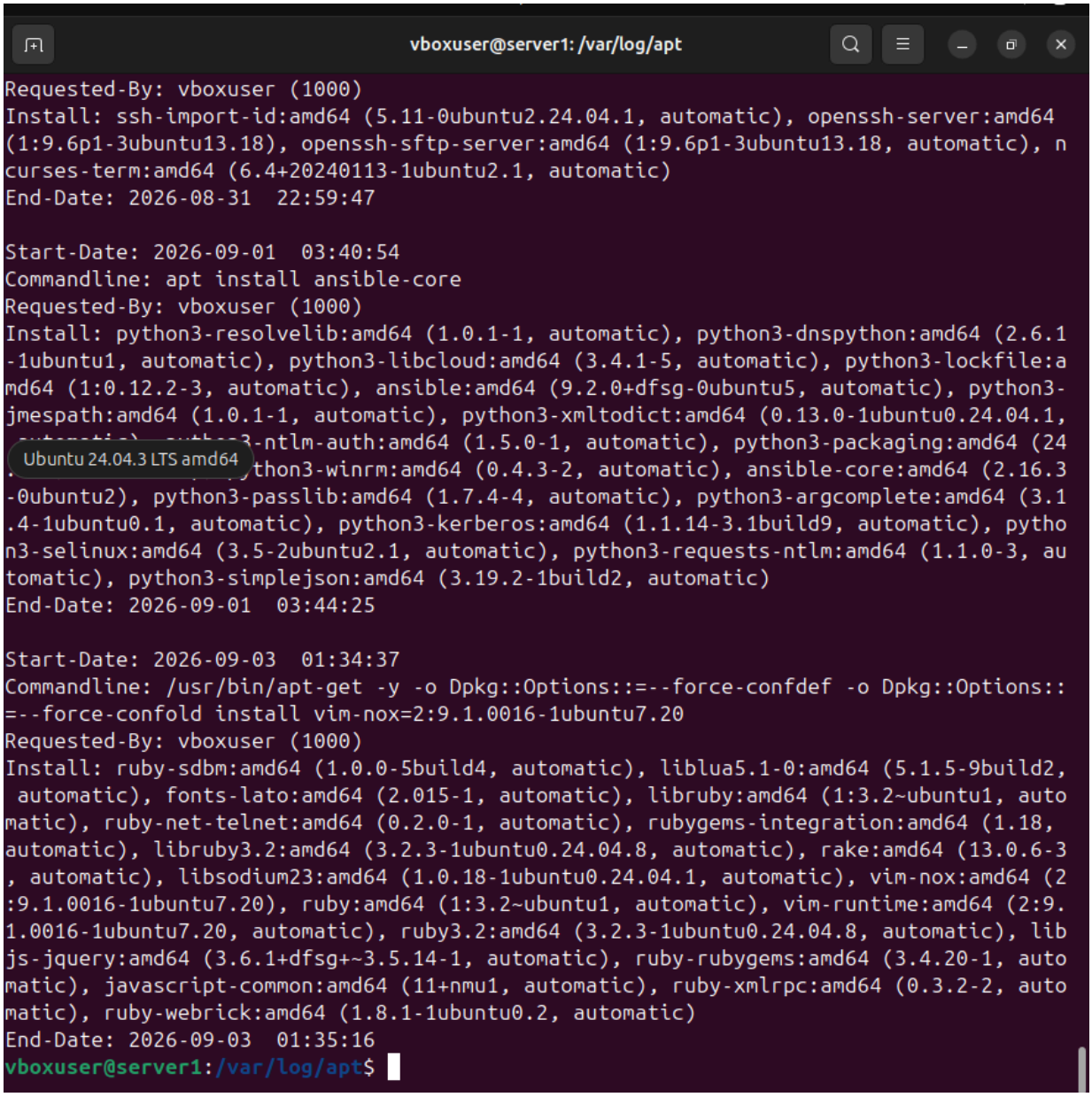
vim-tiny/noble-updates,noble-security,now 2:9.1.0016-1ubuntu7.20 amd64 [installed,autom
atic]
  Vi IMproved - enhanced vi editor - compact version

vboxuser@server1:~/Desktop$ cd /var/log
vboxuser@server1:/var/log$ ls
alternatives.log       cups          dpkg.log         lastlog              vboxadd-inst
all.log
apport.log             cups-browsed  faillog          openvpn              vboxadd-setu
p.log
apt                    dist-upgrade  fontconfig.log   private              vboxadd-setu
p.log.1
auth.log               dmesg         gdm3             README               vboxadd-setu
p.log.2
boot.log               dmesg.0       gpu-manager.log  speech-dispatcher    vboxadd-setu
p.log.3
bootstrap.log          dmesg.1.gz    hp               sssd                 vboxadd-setu
p.log.4
btmp                   dmesg.2.gz    installer        syslog               vboxpostinst
all.log
cloud-init.log         dmesg.3.gz    journal          sysstat              wtmp
cloud-init-output.log  dmesg.4.gz    kern.log         unattended-upgrades
vboxuser@server1:/var/log$ cd apt
```

```
cloud-init-output.log  dmesg.4.gz    kern.log        unattended-upgrades
vboxuser@server1:/var/log$ cd apt
vboxuser@server1:/var/log/apt$ cat history.log

Start-Date: 2025-08-05  16:48:47
Commandline: apt-get -y --fix-policy install
Install: libgpg-error-l10n:amd64 (1.47-3build2, automatic), e2fsprogs-l10n:amd64 (1.47.
0-2.4~exp1ubuntu4, automatic), libgpm2:amd64 (1.20.7-11, automatic), psmisc:amd64 (23.7
-1build1, automatic), uuid-runtime:amd64 (2.39.3-9ubuntu6, automatic), bash-completion:
amd64 (1:2.11-8, automatic), bsdextrautils:amd64 (2.39.3-9ubuntu6, automatic)
End-Date: 2025-08-05  16:48:49

Start-Date: 2025-08-05  16:49:12
Commandline: apt-get --yes -oDebug::pkgDepCache::AutoInstall=yes --force-yes upgrade
Upgrade: libgpg-error-l10n:amd64 (1.47-3build2, 1.47-3build2.1), dpkg:amd64 (1.22.6ubun
tu6, 1.22.6ubuntu6.1), libsmartcols1:amd64 (2.39.3-9ubuntu6, 2.39.3-9ubuntu6.3), libpam
-runtime:amd64 (1.5.3-5ubuntu5, 1.5.3-5ubuntu5.4), udev:amd64 (255.4-1ubuntu8, 255.4-1u
buntu8.10), krb5-locales:amd64 (1.20.1-6ubuntu2, 1.20.1-6ubuntu2.6), python3.12:amd64 (
3.12.3-1, 3.12.3-1ubuntu0.7), libgssapi-krb5-2:amd64 (1.20.1-6ubuntu2, 1.20.1-6ubuntu2.
6), dhcpcd-base:amd64 (1:10.0.6-1ubuntu3, 1:10.0.6-1ubuntu3.1), iputils-ping:amd64 (3:2
0240117-1build1, 3:20240117-1ubuntu0.1), libaudit-common:amd64 (1:3.1.2-2.1build1, 1:3.
1.2-2.1build1.1), apt:amd64 (2.7.14build2, 2.8.3), xkb-data:amd64 (2.41-2ubuntu1, 2.41-
2ubuntu1.1), dbus-user-session:amd64 (1.14.10-4ubuntu4, 1.14.10-4ubuntu4.1), libacl1:am
d64 (2.3.2-1build1, 2.3.2-1build1.1), systemd-timesyncd:amd64 (255.4-1ubuntu8, 255.4-1u
buntu8.10), tzdata:amd64 (2024a-2ubuntu1, 2025b-0ubuntu0.24.04.1), libunistring5:amd64
(1.1-2build1, 1.1-2build1.1), libidn2-0:amd64 (2.3.7-2build1, 2.3.7-2build1.1), libtasn
1-6:amd64 (4.19.0-3build1, 4.19.0-3ubuntu0.24.04.1), python3-minimal:amd64 (3.12.3-0ubu
ntu1, 3.12.3-0ubuntu2), libpam-systemd:amd64 (255.4-1ubuntu8, 255.4-1ubuntu8.10), libse
linux1:amd64 (3.5-2ubuntu2, 3.5-2ubuntu2.1), liblzma5:amd64 (5.6.1+really5.4.5-1, 5.6.1
+really5.4.5-1ubuntu0.2), libhogweed6t64:amd64 (3.9.1-2.2build1, 3.9.1-2.2build1.1), e2
fsprogs-l10n:amd64 (1.47.0-2.4~exp1ubuntu4, 1.47.0-2.4~exp1ubuntu4.1), libcap2-bin:amd6
4 (1:2.66-5ubuntu2, 1:2.66-5ubuntu2.2), gir1.2-glib-2.0:amd64 (2.80.0-6ubuntu1, 2.80.0-
6ubuntu3.4), libext2fs2t64:amd64 (1.47.0-2.4~exp1ubuntu4, 1.47.0-2.4~exp1ubuntu4.1), xx
d:amd64 (2:9.1.0016-1ubuntu7, 2:9.1.0016-1ubuntu7.8), libexpat1:amd64 (2.6.1-2build1, 2
.6.1-2ubuntu0.3), libp11-kit0:amd64 (0.25.3-4ubuntu2, 0.25.3-4ubuntu2.1), libpython3.12
-minimal:amd64 (3.12.3-1, 3.12.3-1ubuntu0.7), kmod:amd64 (31+20240202-2ubuntu7, 31+2024
0202-2ubuntu7.1), dbus-daemon:amd64 (1.14.10-4ubuntu4, 1.14.10-4ubuntu4.1), vim-common:
```



TASK 2

2. ANSWERS TO SUPPLEMENTAL ACTIVITY

Copy and paste the questions from the lab and include your answers after.

Answer the following:
1. What is the importance of using a playbook?

2. Summarize what we have done on this activity.

Conclusions/Learnings:

3. ARTIFICIAL INTELLIGENCE (AI) USE DISCLOSURE STATEMENT

I hereby declare that artificial intelligence (AI) tools were used, if applicable, in the preparation of this academic work. The following indicates the nature and extent of AI assistance:

Tools Used: _____

Examples: ChatGPT, Microsoft Copilot, Grammarly, QuillBot, etc.

Purpose of Use: _____

Examples: grammar correction, idea generation, formatting, summarization, code assistance, explanation of concepts, etc.

Extent of Use: _____

Explain how AI was used and confirm that it did not replace original thinking, analysis, authorship, or completion of the required work.

By submitting this activity, I affirm that my use of AI complies with T.I.P.'s AI usage policy and does not exceed the permitted limits for similarity, automated content generation, or academic assistance.

I understand that any false, incomplete, or misleading disclosure may be subject to investigation in accordance with due process and may result in sanctions for violation of existing T.I.P. policies.

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